MATH 6660: Homework 4
Fall 2026

2D Unstructured FEM:

1. Download the PyDistMesh mesh generator. You can find all the necessary links from the Math 6660 Canvas page.

2. Use this code to generate a mesh for the following geometry: a star-shaped region. The boundary of this region is the zero contour of the following function:

$$\phi(r, \theta) = r - 0.75 - 0.25 \sin(5\theta).$$

3. Use the CG(1) method developed in class and provided in the Math 6660 Canvas page to solve the following Poisson equation on the star-shaped domain created in Problem #2:

$$\begin{aligned} \textbf{PDE}: &\quad -u_{,x,x} - u_{,y,y} = 1 \quad \text{in} \quad \Omega, \\ \textbf{BCs}: &\quad u = 0 \quad \text{on} \quad \partial\Omega. \end{aligned}$$

2D Cartesian FEM:

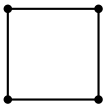
4. Consider the 2D Poisson equation:

$$\begin{aligned} \textbf{PDE}: &\quad -\nabla \cdot \nabla u = f(x, y) \quad \text{in} \quad \Omega = [-1, 1] \times [-1, 1], \\ \textbf{BC}: &\quad u = g \quad \text{on} \quad \partial\Omega. \end{aligned}$$

   (a) Recast this problem as a variational problem. Clearly state the test and trial function spaces.

   (b) Show that the variational problem has a unique solution by showing that it meets all of the criteria of the Lax-Milgram theorem.

   (c) Develop a finite element method for this problem based on square (not triangular) elements of the form:



   On each element, the solution on the four corners is interpolated with the following interpolant:

$$s(x, y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 xy.$$

   Write out the basis functions and explicitly write out the linear system that arises from your FEM (you can just assume that the grid spacing in x and y are the same and uniform).

   (d) Use this method to solve the 2D Poisson equation with

$$\begin{aligned} &f(x, y) = \pi^2 e^{\sin(\pi x)\sin(\pi y)} \left\{ 2\sin(\pi x)\sin(\pi y) + 2\cos^2(\pi x)\cos^2(\pi y) - \cos^2(\pi x) - \cos^2(\pi y) \right\}, \\ &g(x, y) = 1. \end{aligned}$$

   Create a convergence table showing both $\|\cdot\|_V$ and $\|\cdot\|_{L^2(\Omega)}$ errors. What order of convergence do you observe? In this case the exact solution is

$$u(x, y) = e^{\sin(\pi x)\sin(\pi y)}.$$